CIS 508 Final Group Project

EMPLOYEE ATTRITION PREDICTION

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AGENDA

KEY TOPICS DISCUSSED IN THIS PRESENTATION

1. Problem & Solution
2. Conceptual Model
3. Data Mining Solution
4. Data
5. Data Exploration
6. Data Preprocessing
7. Modeling
8. Evaluation Metrics
9. Evaluation
10. Final Model Selection
11. Deployment Plan

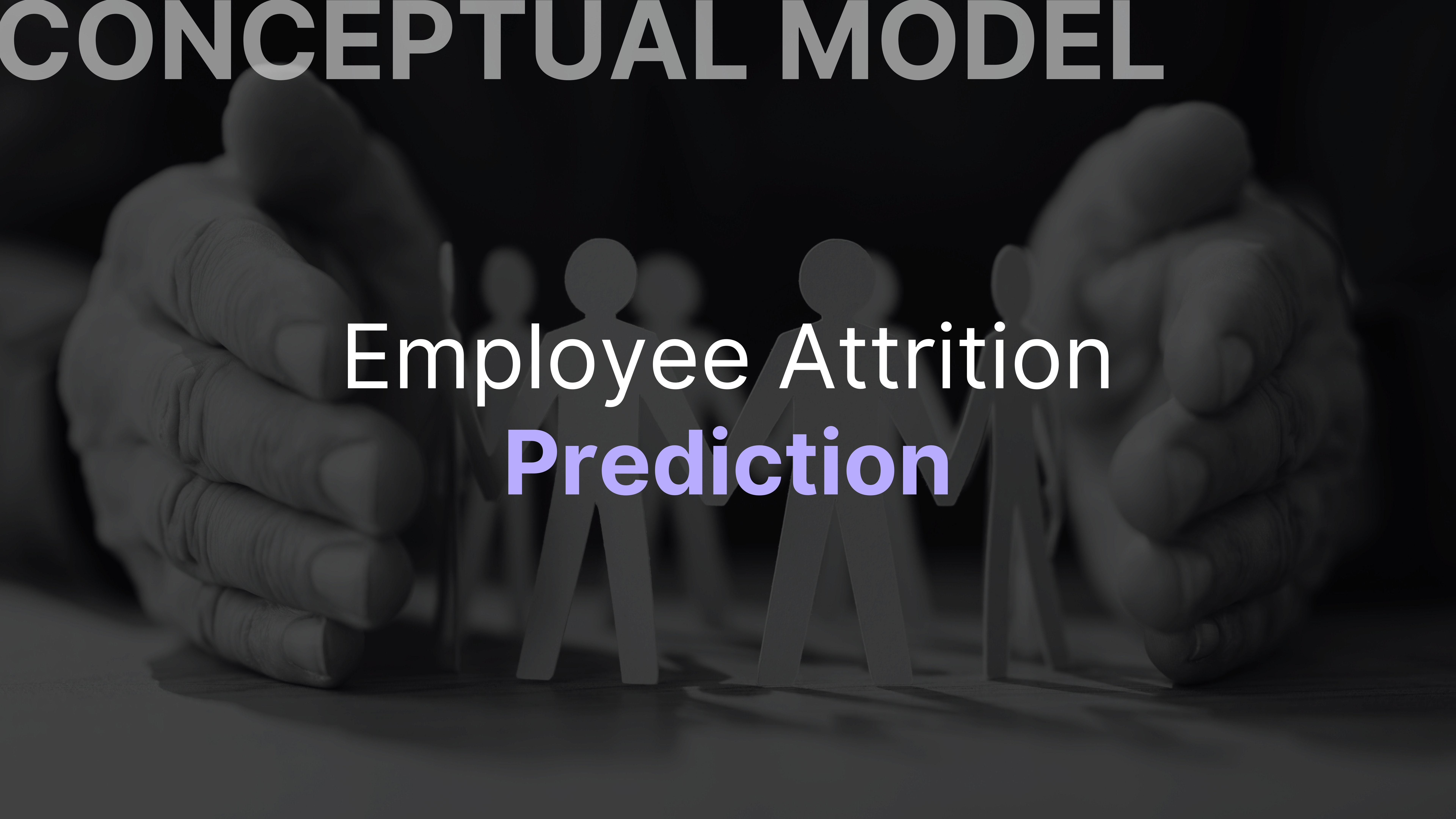
PROBLEM

Employee turnover is a major challenge for many organizations.

High attrition rates can lead to increased recruitment costs, loss of institutional knowledge, and decreased morale.

SOLUTION

By predicting and understanding employee attrition, companies can devise and implement strategies to retain their valuable talent.



CONCEPTUAL MODEL

Employee Attrition

Prediction

DATA MINING SOLUTION

OBJECTIVE

Predict whether a given employees will leave the company

APPROACH

Classification using **supervised** learning

DATA

Data source found in Kaggle

DATASET

Provides a detailed examination of employee attrition and its contributing factors, providing a robust set of quantitative and qualitative data

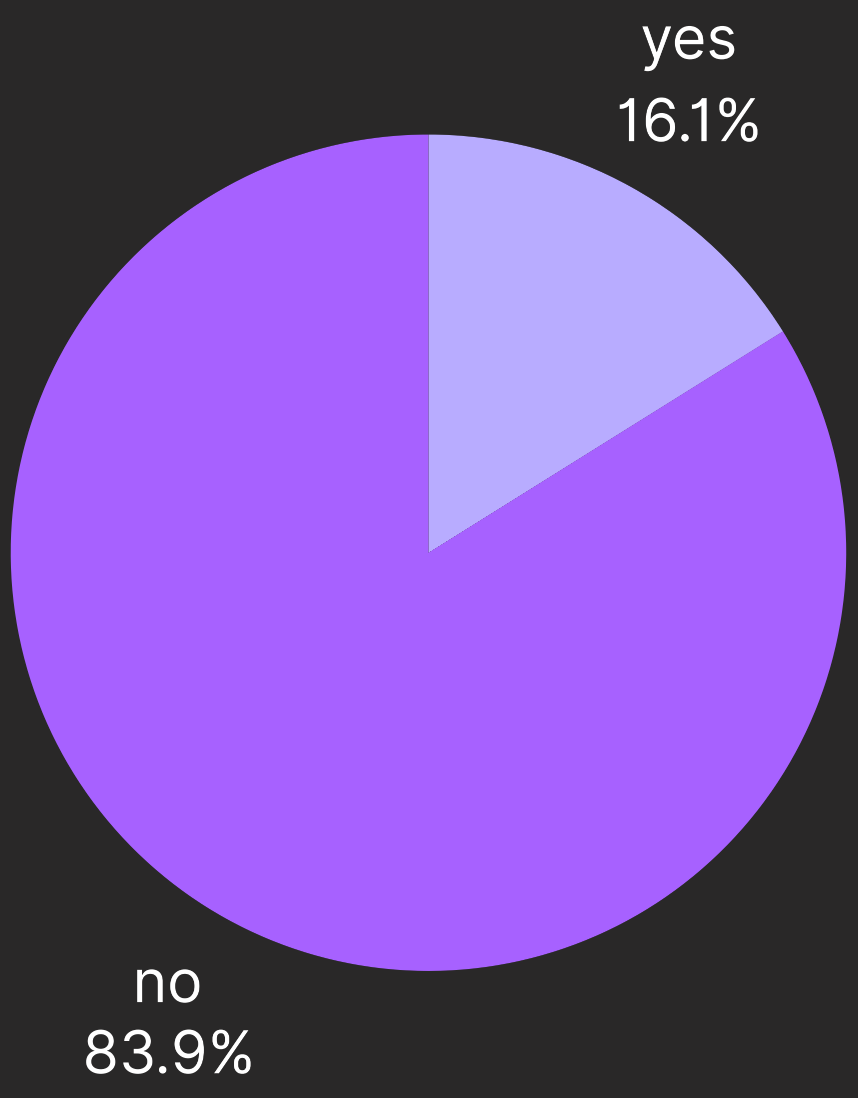
Columns: **35** / Rows: **1470**

Age	Attrition	Business Travel	DailyRate	Department	Distance From Home	Education	...	Worklife Balance	YearsAt Company
41	**Yes**	Travel_Rarely	1102	Sales	1	2	...	1	6
49	**No**	Travel_Frequently	279	Research & Devleopment	8	1	...	3	10
37	**Yes**	Travel_Rarely	1373	Research & Development	2	2	...	3	0
33	**No**	Travel_Frequently	1392	Research & Development	3	4	...	3	8
27	**No**	Travel_Rarely	591	Research & Development	2	1	...	3	2
...	**...**	...	...	...	...	...	...	...	...

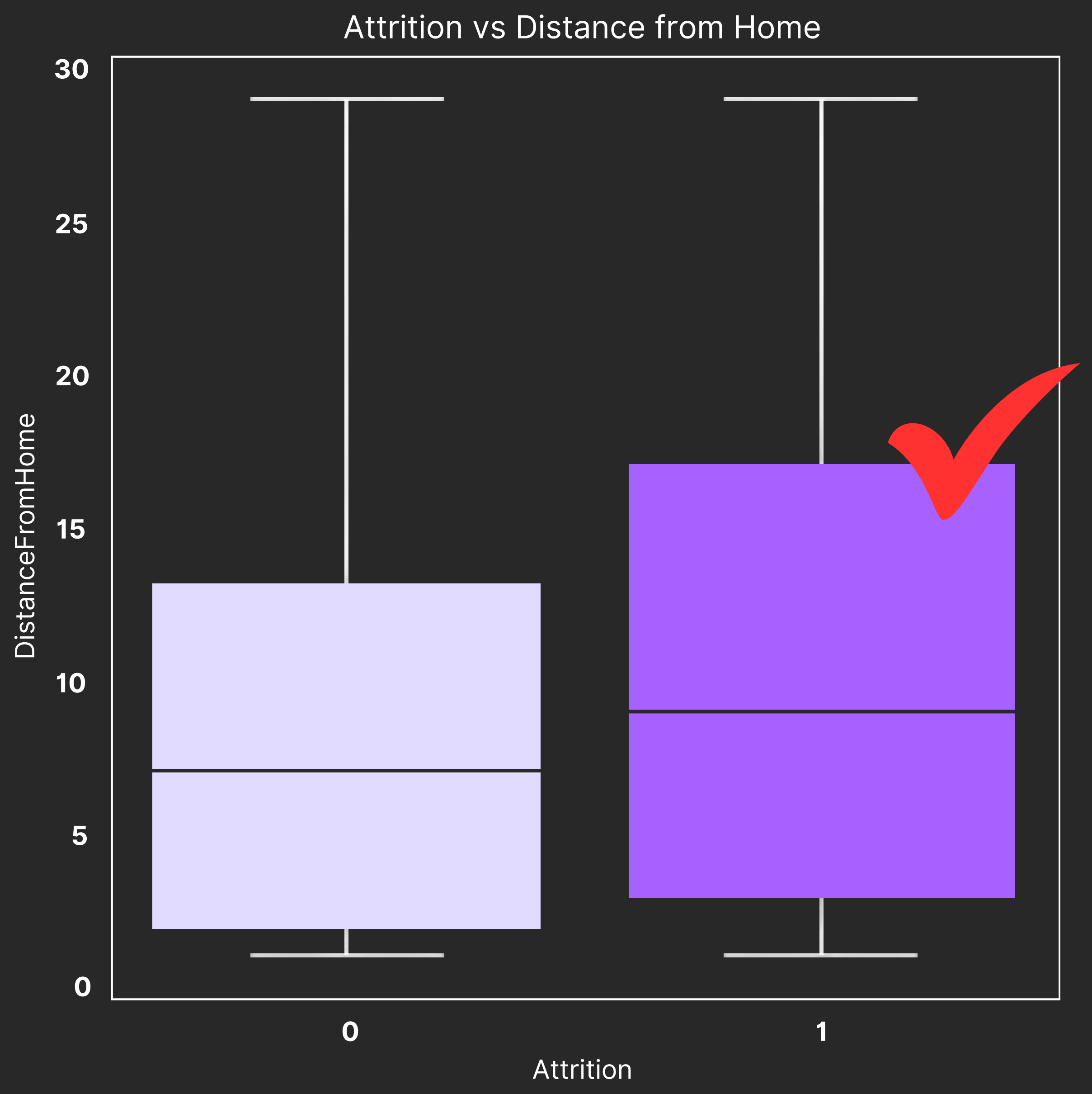
TARGET VARIABLE

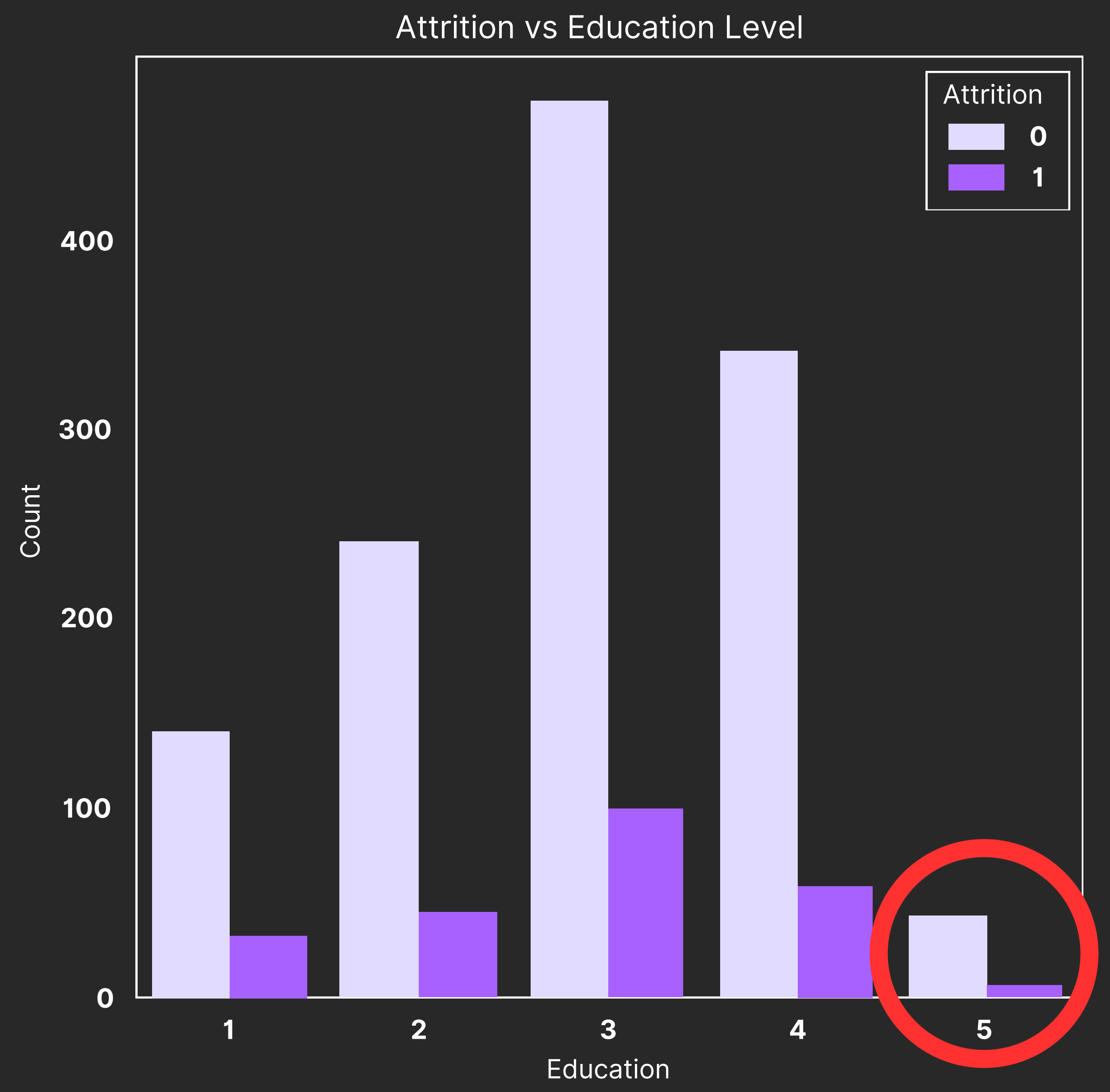
Attrition: Yes/No (binary)

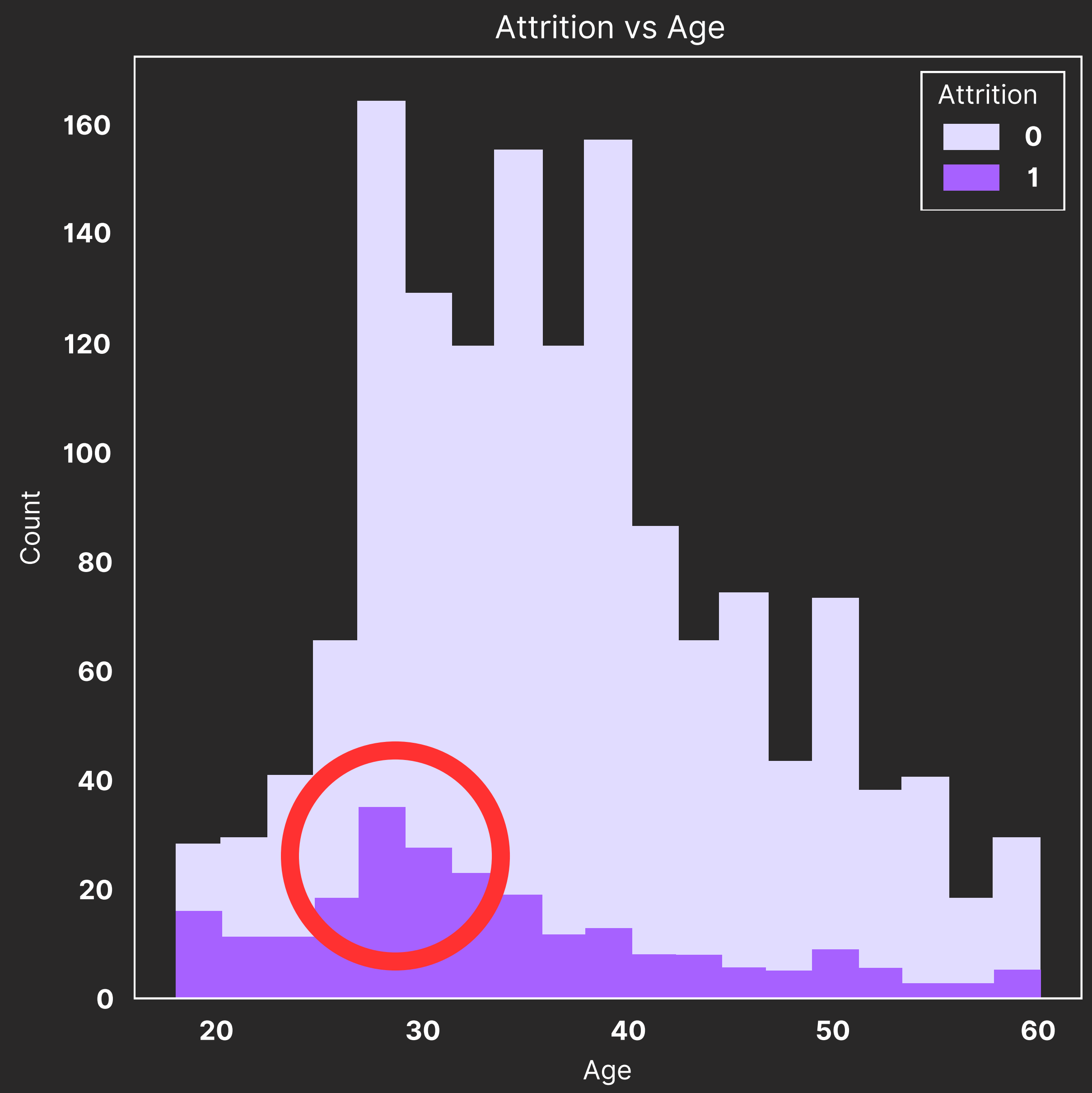
Whether or not the employee has left the organization (categorical)



DATA EXPLORATION





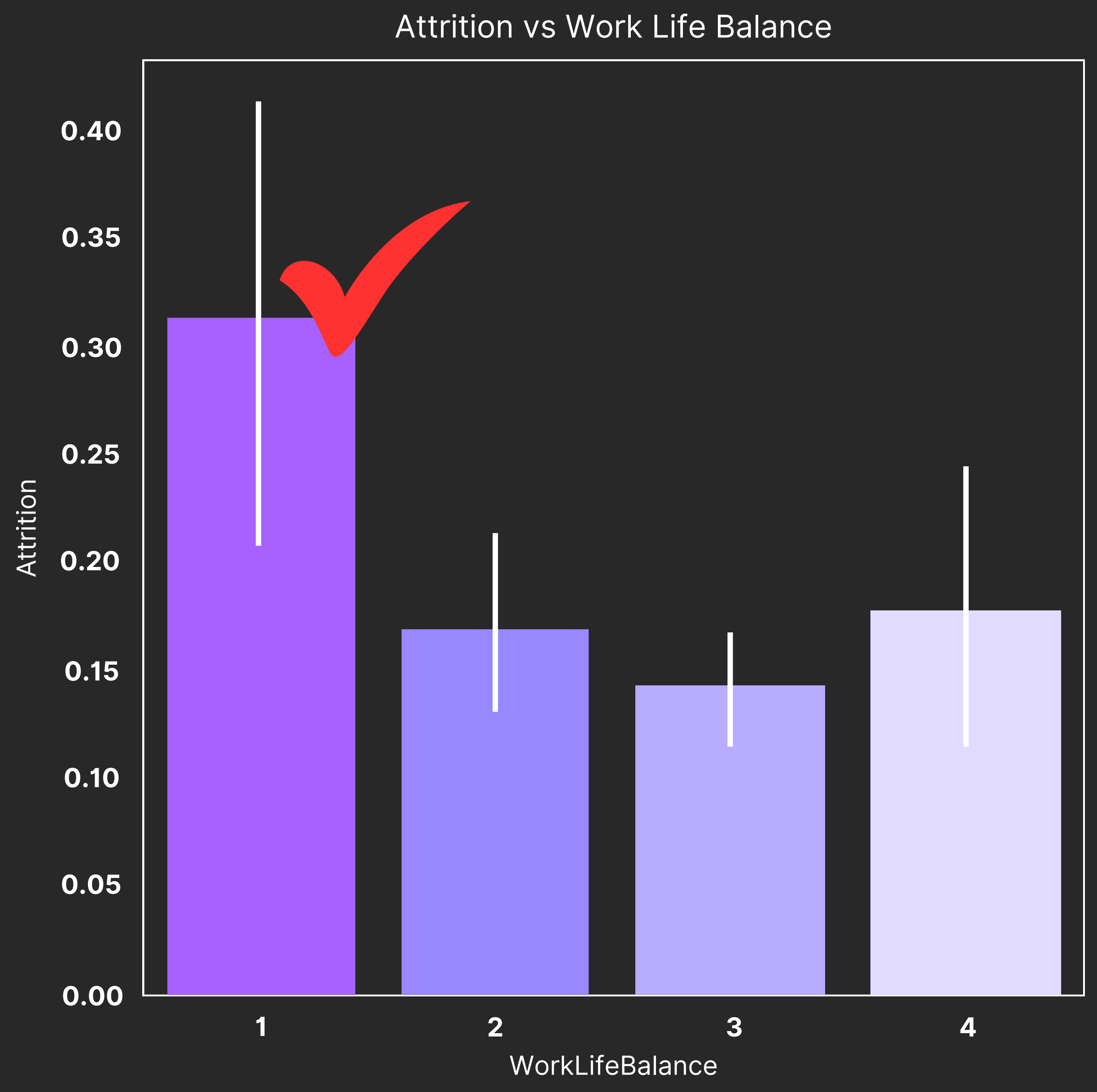


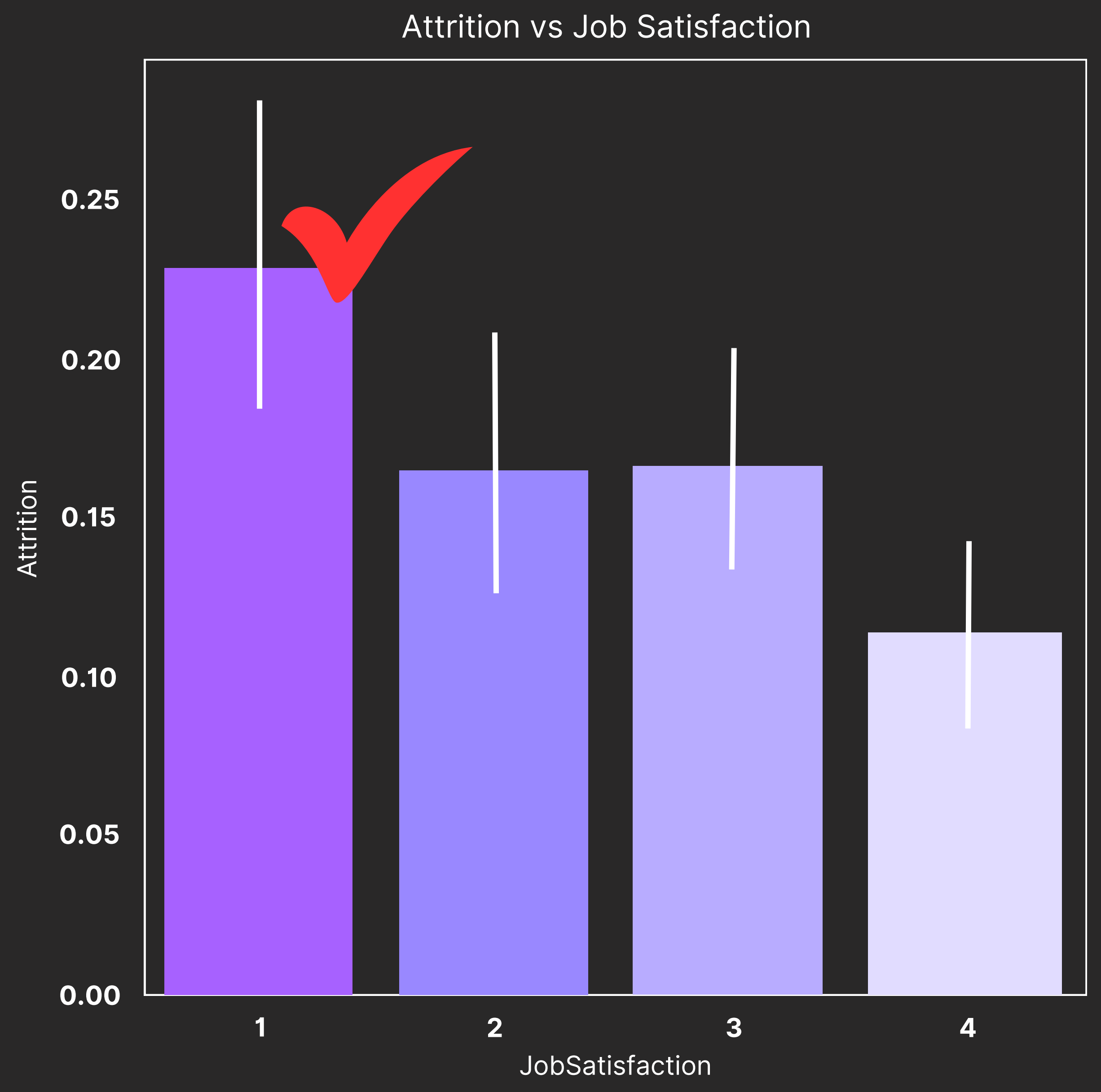
DISTANCE FROM HOME ↑
ATTRITION ↑

EDUCATION LEVEL ↑
ATTRITION ↓

AGE ↓
ATTRITION ↑

DATA EXPLORATION

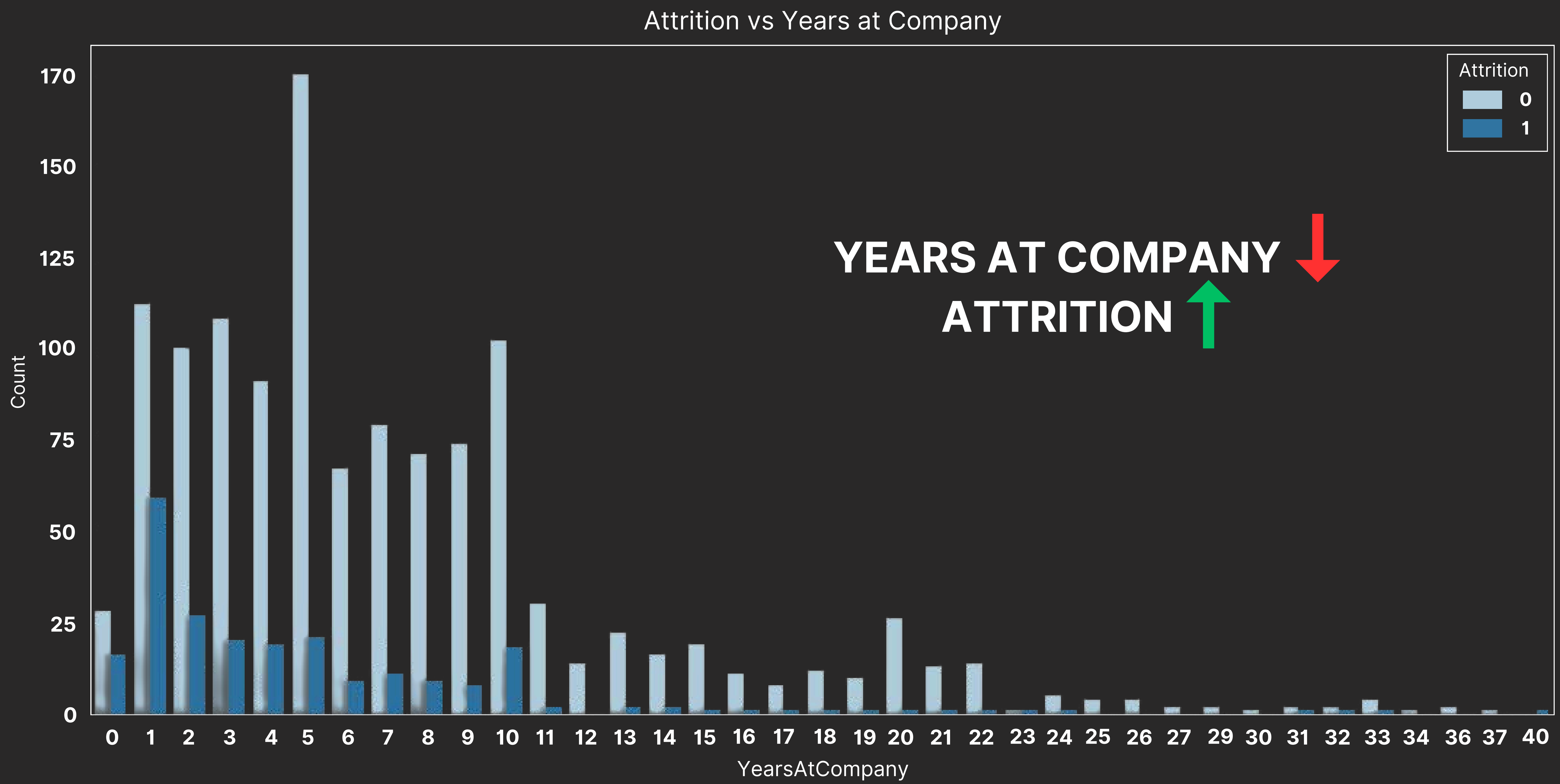




WORK-LIFE BALANCE ↓
ATTRITION ↑

JOB SATISFACTION ↓
ATTRITION ↑

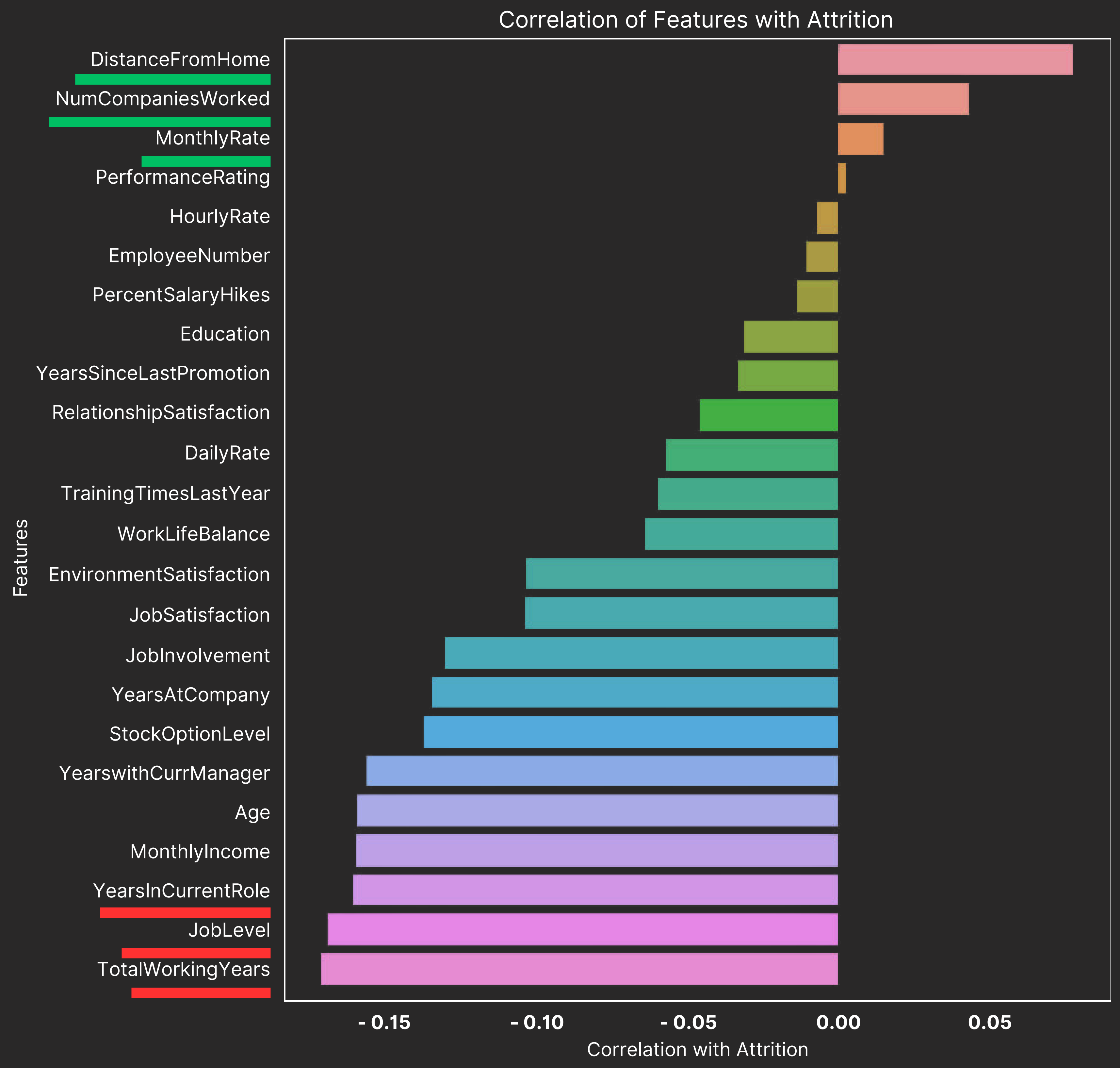
DATA EXPLORATION



YEARS AT COMPANY ↓

ATTRITION ↑

DATA EXPLORATION



1. **Top 3 Highest Positive Correlations:**
   - DistanceFromHome
   - NumCompaniesWorked
   - MonthlyRate

2. **Top 3 Highest Negative Correlations:**
   - TotalWorkingYears
   - JobLevel
   - YearsInCurrentRole

DATA PREPROCESSING

0

ORIGINAL DATA

Age	Attrition	MartialStatus
41	Yes	Single
37	No	Married
33	Yes	Single

1

DATA ENCODING

Age	Attrition	MaritalStatus_Married	MaritalStatus_Single
41	1	0	1
37	0	1	0
33	1	0	1

2

FEATURE SCAILING

Age	Attrition	MaritalStatus_Married	MaritalStatus_Single
0.446350	1	0	1
1.322365	0	1	0
0.008343	1	0	1

3

DATA BALANCING

Count of Attrition
1223
986

→

Count of Attrition
1223
237

4

DATA PARTITIONING

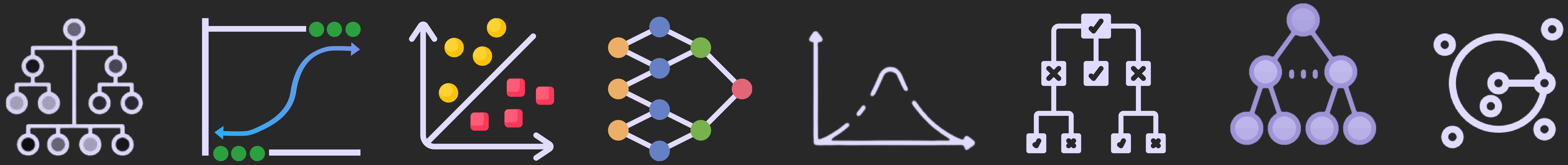
X_train, X_test, y_train, y_test

test_size=0.2, random_state=42

MODELING

17

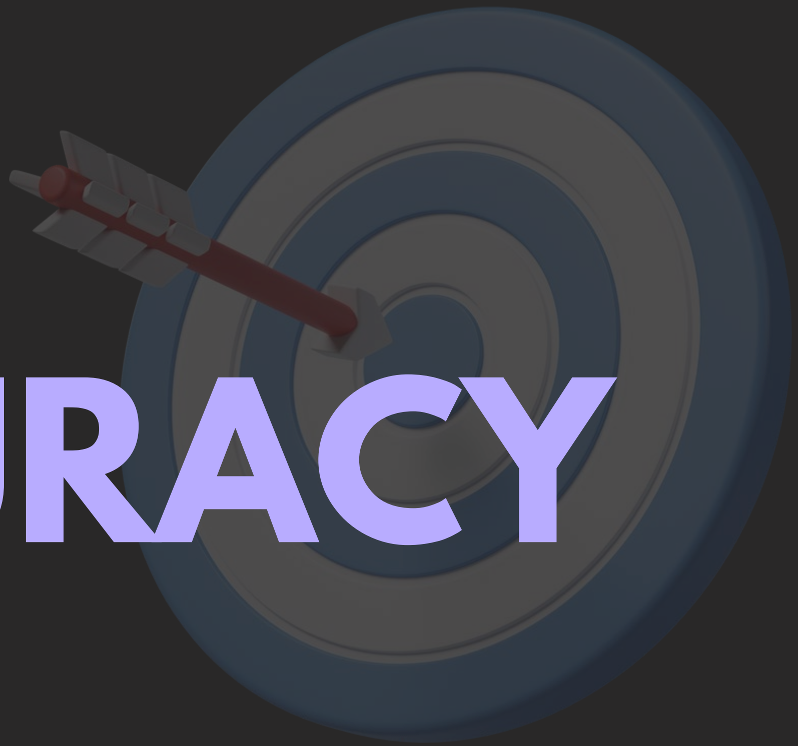
SUPERVISED LEARNING MODEL



- DECISION TREE *
- LOGISTIC REGRESSION *
- SVM *
- NEURAL NETWORKS *
- NAIVE BAYES
- RANDOM FOREST *, ENSEMBLE MODEL
- GRADIENT BOOSTING *, XGBOOSTING *
- KNN *

*: includes hyper-parameter tuned model

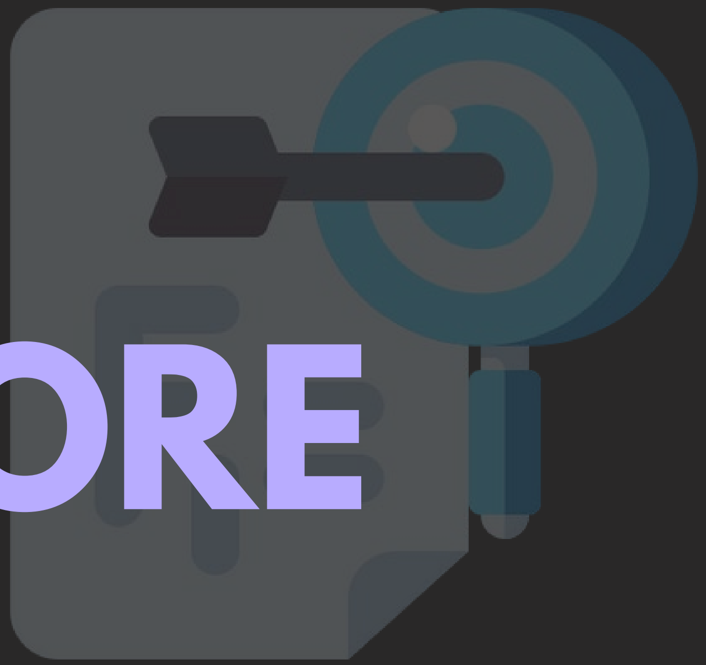
EVALUATION METRICS



ACCURACY

Performance

- Measures correct predictions of **both** staying and leaving employees
- **General performance** of our attrition model



F1-SCORE

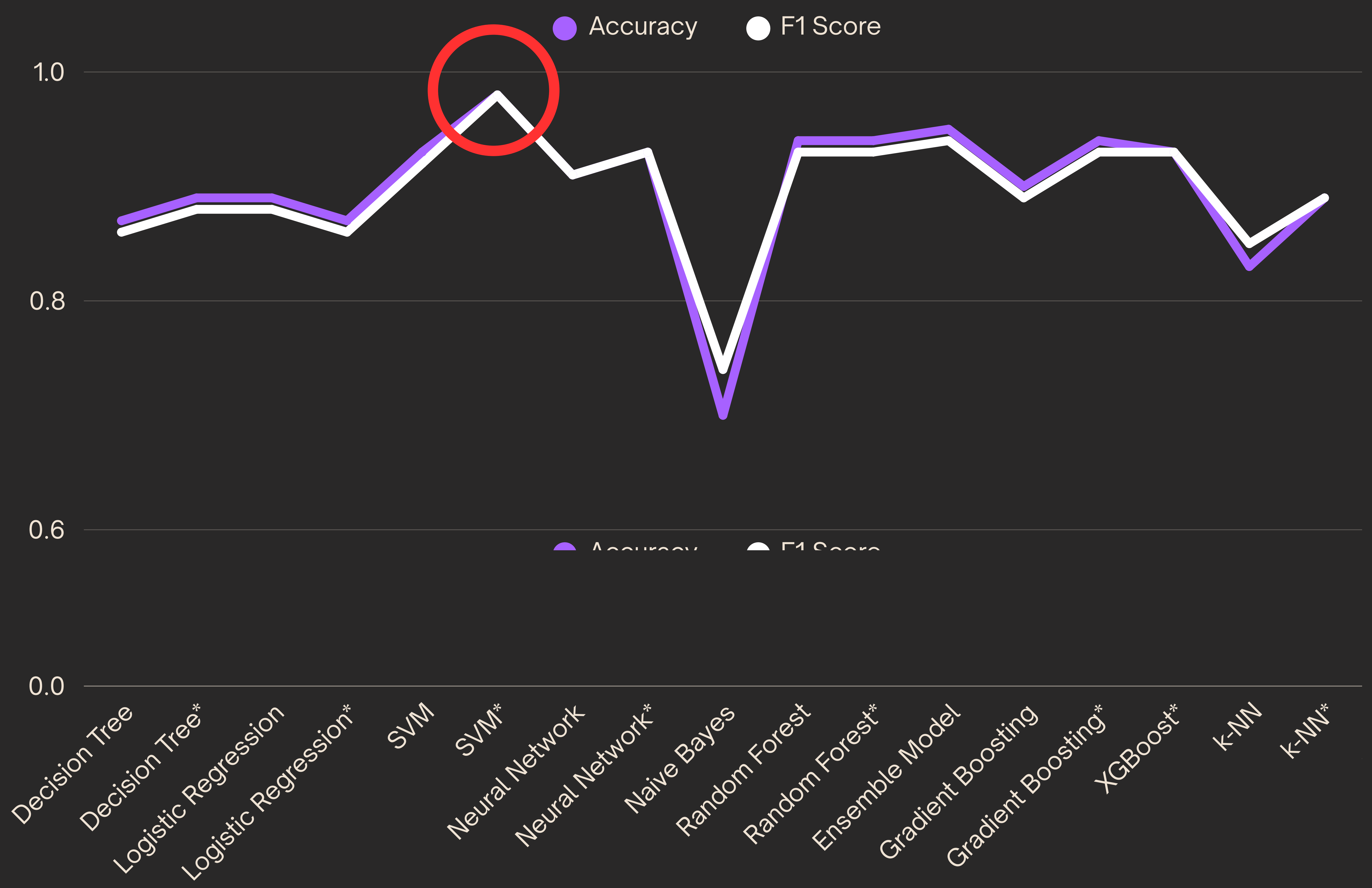
Balance

- Harmonic mean
- **precision** (avoiding false attrition alarms) & **recall** (catching true cases of attrition)

Prediction

- Focuses on the accuracy of predicting the **minority class** (employees likely to leave)

EVALUATION



SVM

with hyperparameter tuned

*: indicates hyper-parameter tuned model

FINAL MODEL

BEST PARAMETERS: {'C': 10, 'gamma': 0.1, 'kernel': 'rbf'}

CLASSIFICATION REPORT:

	Precision	Recall	f1-score	support
0	0.97	0.98	0.98	235
1	0.98	0.97	**0.98**	209
accuracy			**0.98**	444
macro avg	0.98	0.98	0.98	444
weighted avg	0.98	0.98	0.98	444

CONFUSION MATRIX :

	Positive	Negative
Positive	**231**	**4**
Negative	**6**	**203**

SVM

with hyperparameter tuned

OUPUT:

	Attrition
0	1
1	0
2	1
3	0
4	0
5	0
6	0

FEATURE IMPORTANCES

TOP 20 BEST PARAMETERS BY RANK:

1	Over Time
2	Stock Option Level
3	Job Level
4	Total Working Years
5	Age
6	Job Satisfaction
7	Daily Rate
8	Monthly Income
9	Employee Number
10	Department_Research & Development

11	Distance From Home
12	Job Role_Research Scientist
13	Years with Current Manager
14	Years in Current Role
15	Hourly Rate
16	Relationship Satisfaction
17	Years at Company
18	Training Times Last Year
19	Monthly Rate
20	Job Involvement

DEPLOYMENT PLAN

MODEL DEVELOPMENT

- Process employee data and finalize the SVM model
- Export model for deployment

INTEGRATION & LAUNCH

- Choose cloud or on-premises platform
- Set up a secure API for model access
- Implement authentication and encryption

MONITORING & IMPACT

- Ensure system scalability & load management
- Implement monitoring, logging, and CI/CD for updates
- Conduct staging tests & provide API documentation

BUSINESS IMPACT

COST EFFICIENCY

By predicting attrition, companies can **reduce hiring and training costs**

TALENT RETENTION

By understanding the reasons behind attrition, HR can implement **strategies to retain valuable employees**

ENHANCED EMPLOYEE EXPERIENCE

By addressing potential reasons for attrition, the **overall employee experience** can be improved

ANY QUESTIONS?

THANK YOU FOR LISTENING!